Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A stationary coil has constant magnetic flux 0.003 Wb through each turn. Does that constant flux produce sustained induced emf? Explain.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: average speed = distance/elapsed time.
Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Marzano check - circle: ① ② ③

03 Look and explain

Sketch your proposed car. Label the battery, switch, motor, driven axle and wheels. Add a theme that will not obstruct moving parts.



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Build the solution

Assign a build lead, safety checker and recorder. List the teacher approval needed before cutting, drilling, soldering and first power.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

Write one measurable design goal and one constraint. Explain how you will measure the goal instead of judging only by appearance.

Marzano check - circle: 1 2 3

06 Choose a method

An ideal series loop has supply 24 V. One component drops 4 V. Find the other component’s drop.

Marzano check - circle: 1 2 3

07 Exit check

Complete: The part that converts electrical energy into rotation is ____. Before a powered test, I will check ____.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Propose a second success criterion that could conflict with speed, such as straight tracking or reliability. Describe how both would be measured.

Marzano check - circle: 1 2 3